كلية الحاسبات والذكاء الإصطناعي
Faculty of Computers & Artificial Intelligence

Helwan University

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[Image Restoration App]

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Abstract

In today's digital era, the quality and clarity of images are paramount across various applications such as medical imaging, surveillance, and photography. However, images frequently suffer from imperfections including missing regions, noise, and blurriness, which degrade their visual fidelity and usability. These imperfections can obscure important details, mislead interpretations, and diminish the overall impact of the images. Addressing these challenges is crucial to maximizing the effectiveness of visual data. To this end, we propose a comprehensive approach utilizing multi-modal image restoration techniques—specifically inpainting, denoising, and deblurring—through an innovative web application. Our Image Restoration Web App leverages advanced models and algorithms to enhance the quality of degraded images. The app integrates three core functionalities: denoising, deblurring, and inpainting, each designed to tackle specific types of image degradation.

Denoising employs sophisticated algorithms to remove unwanted noise from images, resulting in cleaner and more visually appealing outputs. This process is essential for improving the clarity of images captured in low-light or high-sensitivity conditions. Deblurring, on the other hand, addresses the common issue of image blurriness, which can occur due to camera shake, motion, or out-of-focus lenses. We utilize the NAFNet Model, which we have enhanced with a Multi-head Transposed Attention module, to restore sharpness and fine details to blurred images, making them crisp and clear. Inpainting is particularly useful for restoring old photographs or images with missing or damaged regions. By applying Microsoft's deep latent space translation model, our app can seamlessly fill in these areas, revitalizing old photographs and making them look whole and undamaged.

To run the web app locally, users need to follow a few straightforward steps. First, they must clone the repository to their local machine and navigate to the project directory. Then, they need to install the necessary dependencies, which are specified in the provided requirements file. Additionally, setting up the inpainting model involves downloading files and following instructions from Microsoft's Bringing Old Photos Back to Life project. Once the environment is set up, users can easily run the web app and access it via their browser. The application is designed to be user-friendly, ensuring that even those with minimal technical expertise can effortlessly restore their images. For users who encounter issues with the provided commands for downloading checkpoints, pre-trained models and checkpoints can be manually downloaded from specified links. This project credits the NAFNet model for its contributions to state-of-the-art image restoration and Microsoft's Bringing Old Photos Back to Life project for its advancements in inpainting old photographs. By combining these powerful techniques, our Image Restoration Web App offers a comprehensive solution for enhancing and restoring the visual quality of various images.

Chapter 1

Introduction

1.1) Motivation

1.2) Objectives or Problem Statement

1.3) Similar Systems

1.4) Document Overview

Introduction

The project aims to develop an image restoration web application that can remove noise and blur from images and also fill the missing parts of the images. The application uses computer vision algorithms to restore the images to their original state.

1.1 Motivation

Image restoration is an important problem in the field of image processing using deep learning algorithms, because it addresses the challenge of recovering or enhancing degraded or corrupted images.

There are several reasons why image restoration is important:

Preservation of Visual Information: Images can be corrupted due to various factors such as noise, blur, or sensor limitations. Image restoration techniques aim to recover the original visual information that may have been lost or distorted during the image acquisition or storage process. By restoring images, valuable visual details can be preserved.

Enhancing Visual Quality: Image restoration techniques can also be used to enhance the visual quality of images by reducing noise, removing unwanted artifacts, or improving the overall sharpness and clarity. This is particularly valuable in applications such as photography, medical imaging, satellite imagery, and surveillance systems, where high-quality images are crucial for accurate analysis, diagnosis, or decision-making.

Historical Document Preservation: Image restoration plays a significant role in preserving and restoring historical documents, photographs, and artworks. These artifacts may deteriorate over time due to physical degradation, fading, or other environmental factors. By applying restoration techniques, the original appearance and content of these valuable cultural assets can be recovered, ensuring their long-term preservation and accessibility.

Forensic Analysis: Image restoration techniques are widely used in forensic analysis and criminal investigations. In forensic imagery, it is often necessary to enhance low-quality or degraded images to extract important details such as faces, license plates, or other evidence. By restoring and enhancing these images, investigators can improve their ability to identify suspects, reconstruct crime scenes, or analyse critical details.

Computer Vision Applications: Image restoration is an essential component in many computer vision applications such as object recognition, image segmentation, and 3D reconstruction. By restoring images to their optimal quality, these downstream tasks can benefit from cleaner and more accurate input data, leading to improved performance and reliability of computer vision algorithms.

Overall, image restoration techniques are crucial for a wide range of domains. By addressing the challenges of image degradation and corruption, these techniques contribute to the advancement of image processing, computer vision, deep learning, and the overall understanding and utilization of visual data.

1.2 Problem Definition

Digital images are a crucial part of our visual culture, used in various fields such as photography, medical imaging, and computer vision. However, images are often affected by noise and blur, which can significantly impact their quality and usability.

Understanding Noise in Images: Noise in images refers to unwanted variations in pixel values that result in a loss of image clarity and detail. It can be caused by various factors, including sensor limitations, low lighting conditions, or image compression. Noise manifests as random speckles or grainy patterns in the image and can be categorized into two main types: luminance noise (grayscale variations) and color noise (color variations)

Effects of Noise:

1-Reduced Image Quality: Noise deteriorates image quality, making it appear less sharp and clear. Fine details can become obscured by noise, especially in low-light or high-ISO images.

2. Loss of Color Fidelity: Color noise can distort the original colors in an image, resulting in color artifacts that affect the overall visual appeal.

Understanding Image Blurriness: Image blurriness occurs when the pixels in an image deviate from their true positions, creating a lack of sharpness. Blur can result from various sources, including camera shake, motion, or optical imperfections. Dealing with blur involves employing techniques that aim to restore image sharpness and recover lost details.

Effects of Blurriness:

- Reduced Image Clarity: Blurriness directly affects the clarity and sharpness of an image, making fine details less discernible. This can result in a loss of visual impact and reduced overall image quality.

2. Loss of Information: In applications where precise details are critical, such as medical imaging or forensic analysis, blurriness can lead to a loss of valuable information, potentially affecting diagnostic accuracy or investigative efforts

Noise and blurriness are ubiquitous challenges in digital imagery, exerting substantial influence on image quality and interpretability. Understanding the ramifications of noise and the benefits of deblurring techniques is vital for photographers, medical practitioners, and anyone working with digital images

1.3 Similar System Information

The researchers have made significant contributions in addressing denoising and deblurring issues in image processing by developing various models such as NAFNet, Restormer, UNet, and plain blocks. They measured the effectiveness of these models using PSNR and SSIM metrics. In a recent paper, they explored the use of non-linear activation functions to enhance the performance of the models. Overall, their efforts have led to significant advancements in image processing, which holds great potential for various applications such as medical imaging and satellite imaging.

The researchers have made several important contributions to the field of image denoising and deblurring. These contributions include:

The development of new models for image denoising and deblurring, such as NAFNet, Restormer, and UNet.

The use of non-linear activation functions to enhance the performance of image denoising models.

The development of new evaluation metrics for image denoising and deblurring, such as the Perceptual Evaluation of Signal Quality (PESQ) metric.

The open-source release of their code and datasets, which has enabled other researchers to build on their work.

Their work has led to the development of new models and techniques that can be used to improve the quality of images. Their work also has the potential to be used in a variety of applications, such as medical imaging and satellite imaging.

1.4 Document Overview

Chapter 1: Introduction

This chapter provides an overview of the development of image restoration methods within the context of deep learning. It highlights the significant performance improvements achieved by deep learning-based approaches, citing specific examples. However, it points out that these methods often suffer from high system complexity, which is further broken down into two parts: inter-block complexity and intra-block complexity.

Chapter 2: Related Work

The chapter discusses various research papers on image restoration, focusing on denoising and deblurring tasks. Notable papers include one introducing a network called DLEFNet for motion deblurring using event cameras. Another paper proposes a computationally efficient method, NafNet, for denoising and deblurring. Additionally, a paper presents RealSR, a framework for real-world super-resolution by estimating blur kernels and injecting noise. Another paper addresses the challenge of restoring old photos with a Triplet Domain Translation Network. Lastly, Restormer introduces an efficient Transformer model for high-resolution image restoration, achieving state-of-the-art results across various tasks. Your selection of NafNet and incorporating attention modules from Restormer reflects a strategic choice based on the papers' advancements in image restoration.

Chapter 3 : System Architecture

In this chapter, we delve into the system architecture of our image restoration web app, which employs the state-of-the-art NafNet model to address the challenges of image deblurring, denoising, and inpainting. We start with a high-level design overview, explaining the rationale behind choosing and modifying the NafNet model. Following this, we detail the specific components and their functionalities within our architecture.

Chapter 4: System Design

In this chapter, we explore the system architecture of the image restoration web app is designed to provide a seamless and efficient user experience for restoring images through deblurring, denoising, and inpainting processes. The architecture is divided into three main phases: User Interface (UI), Backend, and Output. Each phase plays a crucial role in ensuring the functionality, performance, and reliability of the application.

Chapter 5 : Proposed system

In This chapter outlines the proposed image restoration system, detailing its design, datasets, performance, tools, and future work. The system leverages the advanced NAFNet model to address tasks such as image deblurring, denoising, and inpainting. The structure of this chapter includes a thorough description of the system, the datasets used for training and evaluation, performance metrics, and the tools and hardware involved. Additionally, it covers future improvements and provides snapshots and code samples illustrating the system's functionality.

Chapter 2

Related Work

1- Deformable Convolutions and LSTM-based Flexible Event Frame Fusion Network for Motion Deblurring [1]

Authors: Dan Yang, Mehmet Yamac

This paper presents a novel approach for motion deblurring that leverages the unique capabilities of event cameras in combination with traditional RGB cameras. Event cameras capture only changes in the scene asynchronously, providing high temporal resolution and useful information for deblurring fast-moving objects. However, integrating this event data with RGB data poses challenges due to its asynchronous nature and the dynamic exposure times of modern cameras. To address these challenges, the authors propose a network named Deformable Convolutions and LSTM-based Flexible Event Frame Fusion Network (DLEFNet). This network incorporates a Long Short-Term Memory (LSTM) module for dynamic event feature extraction and deformable convolutions to enhance feature extraction from event data. Unlike previous methods that use a fixed number of event frames, DLEFNet adapts to the variable number of event frames dictated by the scene's dynamics and the camera's exposure time.

Key Contributions

1- Dynamic Event Feature Extraction Module: Utilizes LSTM to handle varying numbers of event frames, improving temporal resolution and adaptability to different exposure times.
2- Deformable Convolutions: Incorporated into the LSTM module to enhance feature extraction by adjusting the receptive fields dynamically.
3- Integration with RGB Frames: The network combines extracted event features with RGB frame features across multiple scales for effective deblurring.
4- Performance: Demonstrated state-of-the-art performance on both synthetic (GoPro dataset) and real-world (REBlur dataset) scenarios, outperforming existing deblurring methods.

Evaluation

The proposed method shows significant improvements in deblurring accuracy, especially in scenarios with varying exposure times and fast-moving objects. The network outperforms other state-of-the-art methods, as evidenced by higher Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) scores in benchmarks.

Conclusion

DLEFNet effectively utilizes the high temporal resolution of event cameras to improve motion deblurring in RGB images. Its flexible approach to handling dynamic event frames and integration with deformable convolutions sets a new benchmark for deblurring performance in both synthetic and real-world datasets.

2- Simple Baselines for Image Restoration (NafNet model)[2]

Author: Liangyu Chen , Xiaojie Chu , Xiangyu Zhang, and Jian Sun

The research paper addresses the challenging problems of denoising and deblurring in image restoration. The authors set out to develop effective and computationally efficient methods for these tasks by leveraging state-of-the-art (SOTA) techniques and proposing their own innovative approach.

To achieve high-quality restoration results, the researchers adopted the UNet architecture, a popular and successful deep learning model for image restoration tasks. They enhanced the UNet architecture by incorporating normalization layers and utilizing activation functions such as RELU and GELU. These modifications helped improve the model's performance and its ability to handle complex image restoration challenges.

In the initial implementation, the proposed method achieved a peak signal-to-noise ratio (PSNR) of 39.85, indicating the restoration of high-quality images. This result demonstrated the efficacy of the approach and its ability to effectively address denoising and deblurring problems.

Building on this success, the researchers further explored the potential of alternative techniques. They investigated the use of NAFNet, a different network architecture, and GLU (Gated Linear Unit) as an activation function. Additionally, they incorporated simplified channel attention, a mechanism that helps the network focus on important image features during the restoration process. This exploration proved to be promising and yielded even more impressive results.

For denoising evaluation, the researchers utilized the SIDD dataset, a widely used benchmark for image denoising tasks. The proposed model achieved an outstanding accuracy of 40.30dB/0.961, demonstrating its effectiveness in removing noise and improving image quality. For deblurring, the researchers employed the GOPRO dataset, which is specifically designed for evaluating image deblurring algorithms. The proposed method achieved an accuracy of 33.69dB/0.967, showcasing its capability to restore sharp and clear images by mitigating the effects of blurring.

One notable aspect of the proposed approach is its computational efficiency. Both denoising and deblurring tasks were accomplished with a total of 65 million multiply-accumulate operations (MACs). This indicates that the method achieves impressive restoration performance while remaining relatively lightweight in terms of computational requirements. Such efficiency makes the proposed approach suitable for real-time applications or scenarios with limited computational resources.

In conclusion, the research paper "Simple Baselines for Image Restoration" presents a novel approach that effectively addresses the challenges of denoising and deblurring in image restoration. By incorporating advanced techniques, such as the UNet architecture with normalization layers and various activation functions, the proposed method achieves high-quality restoration results. The promising exploration of alternative techniques and the impressive accuracies obtained on benchmark datasets further validate the effectiveness of the approach. Additionally, the computational efficiency demonstrated by the model makes it a practical solution for real-world applications.

3- Real-World Super-Resolution via Kernel Estimation and Noise Injection [3]

Author: Xiaozhong Ji , Yun Cao , Ying Tai , Chengjie Wang , Jilin Li , Feiyue Huang , Nanjing University, Tencent Youtu Lab

Recent state-of-the-art super-resolution methods have excelled on ideal datasets, but struggle with real-world images due to simplistic bicubic downsampling used during training. This paper introduces a novel degradation framework for real-world images by estimating blur kernels and real noise distributions. This framework creates low-resolution images that better mimic real-world conditions. The proposed real-world super-resolution model outperforms current methods, achieving lower noise and better visual quality, and won the NTIRE 2020 Challenge on Real-World Super-Resolution.

1- Introduction
Super-resolution (SR) increases image resolution and clarity. Recent deep learning-based methods have shown significant results on datasets downsampled using fixed bicubic kernels. However, these methods perform poorly on real-world images as they fail to account for the frequency-related details lost in bicubic downsampling. Real-world images require accurate degradation modeling to ensure the generated low-resolution (LR) image matches the domain attributes of real-world images.

2- Related Work Super-Resolution: Convolutional Neural Networks (CNN)-based methods like EDSR achieve strong performance on bicubic downsampled images, focusing mainly on fidelity. Generative Adversarial Networks (GAN)-based methods enhance visual effects by introducing adversarial and perceptual losses. However, these models struggle with real-world images due to their training on clean, bicubic downsampled data. Real-World Super-Resolution: Recent methods integrate denoising or deblurring but require sufficient prior knowledge about blur/noise, limiting their application. Real-world super-resolution challenges have prompted novel methods, such as DSGAN for generating degraded images and unsupervised learning approaches. However, accurate kernel estimation for degradation modeling is crucial for producing clear and sharp results.

3- The Proposed RealSR The RealSR framework consists of two main stages: realistic degradation for super-resolution and training the SR model.

3.1 Realistic Degradation for Super-Resolution The degradation process combines kernel estimation and noise injection: Kernel Estimation and Downsampling: KernelGAN is adapted to estimate blur kernels, preserving low-frequency information from the source image. Noise Injection: Noise patches are collected from the source dataset and added to downsampled images to simulate real-world conditions.

3.2 Super-Resolution Model An SR model based on ESRGAN, using RRDB structures, is trained on the constructed data pairs. The training incorporates pixel loss, perceptual loss, and adversarial loss to optimize both fidelity and visual quality.

3.3 Patch Discriminator in RealSR A patch discriminator replaces VGG-128, focusing on local details to minimize artifacts and improve local feature representation.

4- Experiments

4.1 Datasets DF2K: A merged dataset from DIV2K and Flickr2K, containing 3,450 images with synthetic Gaussian noise. DPED: Contains 5,614 real-world images from the iPhone3 camera, featuring natural noise and blur.

4.2 Evaluation Metrics PSNR and SSIM: Common metrics for image restoration, emphasizing fidelity. LPIPS: Evaluates visual quality by comparing perceptual features.

4.3 Evaluation on Corrupted Images The RealSR method outperforms EDSR, ESRGAN, ZSSR, and K-ZSSR on the DF2K dataset, achieving the best LPIPS score, indicating superior visual quality.

4.4 Evaluation on Real-World Images Visual comparisons on the DPED dataset demonstrate RealSR's ability to produce clearer images with fewer artifacts compared to other methods.

Conclusion

The proposed RealSR method provides a novel degradation framework for real-world images, significantly improving visual quality in super-resolution tasks. Extensive experiments demonstrate its superiority over state-of-the-art methods, particularly in real-world applications. The RealSR model effectively handles noise and blur, ensuring high-quality SR results suitable for practical use.

4- Bringing Old Photos Back to Life [4]

Author: Ziyu Wan, Bo Zhang, Dongdong Chen, Pan Zhang, Dong Chen, Jing Liao, and Fang Wen

Overview: The paper addresses the restoration of old photos suffering from severe degradation. Unlike conventional restoration tasks, real photos exhibit complex degradation, and the domain gap between synthetic and real images poses challenges for generalization.

Key Contributions:

1. **Triplet Domain Translation Network**: The authors propose a novel network that leverages real photos and synthetic image pairs. Two variational autoencoders (VAEs) transform old and clean photos into latent spaces, bridging the domain gap effectively.
2. **Handling Multiple Degradations**: The method includes a global branch for structured defects (scratches, dust spots) and a local branch for unstructured defects (noises, blurriness). These branches are fused in the latent space.
3. **Visual Quality**: The proposed approach outperforms existing methods in restoring old photos.

Evaluation: The paper demonstrates improved visual quality and robustness across various degradation types.

Conclusion: The authors provide an effective solution for restoring old photos, considering real-world complexities. Their approach bridges the gap between synthetic and real images, resulting in high-quality restoration.

5- **Restormer: Efficient Transformer for High-Resolution Image Restoration** [5]

Authors: Syed Waqas Zamir, Aditya Arora, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, Ming-Hsuan Yang

Overview:

- Image restoration involves reconstructing high-quality images by removing degradations (such as noise, blur, and raindrops) from degraded inputs.
- Convolutional neural networks (CNNs) are commonly used for learning image priors from large-scale data. However, they have limitations, including a limited receptive field and inadaptability to input content.
- Transformers, another class of neural architectures, have shown significant performance gains in natural language and high-level vision tasks. However, their computational complexity grows quadratically with spatial resolution, making them infeasible for most high-resolution image restoration tasks.
- The authors propose an efficient Transformer model called "Restormer" by designing key components (multi-head attention and feed-forward network) to capture long-range pixel interactions while remaining applicable to large images.

- **Key Contributions**:
  - Efficient Transformer architecture for image restoration.
  - State-of-the-art results on various restoration tasks, including deraining, single-image motion deblurring, defocus deblurring (single-image and dual-pixel data), and image denoising (Gaussian grayscale/color denoising and real image denoising).
- **Evaluation**:
  - The evaluation section in a research paper provides an assessment of the proposed method's performance. It involves comparing the model's results to existing methods or baselines using various metrics.
  - In the case of "Restormer," the authors evaluate their efficient Transformer model on several image restoration tasks, including:
    - **Image Deraining**: Removing raindrops from images.
    - **Single-Image Motion Deblurring**: Reducing blur caused by camera or object motion.
    - **Defocus Deblurring**:
      - **Single-Image Defocus Deblurring**: Correcting defocus blur due to incorrect camera focus.
      - **Dual-Pixel Defocus Deblurring**: Handling defocus blur using dual-pixel data.
    - **Image Denoising**:
      - **Gaussian Grayscale/Color Denoising**: Reducing noise in grayscale or color images.
      - **Real Image Denoising**: Handling noise in real-world images.

    - The evaluation metrics used may include Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), or other relevant measures specific to each task.
    - By comparing Restormer's results with state-of-the-art approaches, the authors demonstrate its effectiveness in image restoration.
- **Conclusion**:
    - The conclusion section summarizes the key findings, contributions, and implications of the research.
    - In the case of "Restormer":
        - **Summary of Contributions**:
            - Introduction of an efficient Transformer model for high-resolution image restoration.
            - State-of-the-art performance on various restoration tasks.
        - **Implications and Significance**:
            - Restormer's ability to handle long-range pixel interactions while remaining applicable to large images is crucial for practical use.
            - The findings advance the field of image restoration by leveraging Transformer architectures.
        - **Future Directions**:
            - Suggestions for further research, potential improvements, or extensions of the proposed model.
        - **Closing Remarks**:
            - Emphasizes the importance of Restormer's contributions to image restoration.

Through those papers, to build a denoising and deblurring model we have chosen" Simple Baselines for Image Restoration (NAFNet Model)" because it achieved the highest results and we have replaced simplified channel attention (SCA) module from NAFNet model with Multi head transposed attention module from the paper "Restormer: Efficient Transformer for High-Resolution Image Restoration" and we have chosen "Bringing Old Photos Back to Life" to build an inpainting model

Chapter 3

System Architecture

3.1) Design Overview

3.2) Chosen System Architecture

3.3) Detailed Description of Components

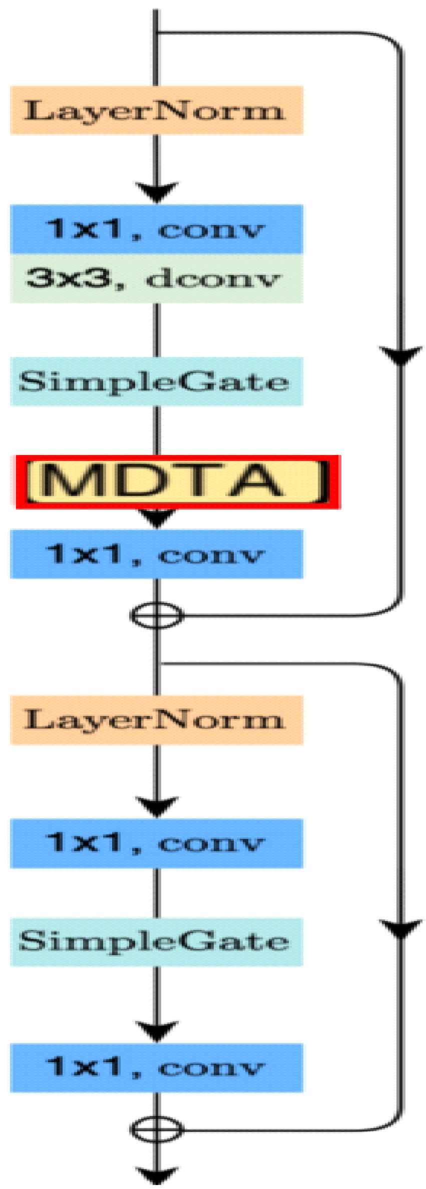
3.4) Inpainting Model

3.1) Design Overview

The original NAFNet model uses a simplified channel attention (SCA) module to learn the importance of each channel in the feature maps. The SCA module is computationally expensive and can be inefficient for feature mapping. And the Multi-dconv Head Transposed Attention (MDTA) module is an efficient way to learn the importance of each channel and it can also improve the feature mapping.

3.2) Chosen System Architecture

Hesham et al. decided to use the NAFNet model to solve the problem of image denoising and deblurring. We decided to change the simplified channel attention (SCA) module with the Multi-dconv Head Transposed Attention (MDTA) to make it more efficient in feature mapping and less computationally expensive.



3.3) Detailed Description of Components

3.3.1 Architecture

To reduce the inter-block complexity, we adopt the classic single-stage U-shaped architecture with skip-connections, as shown following [39,36].

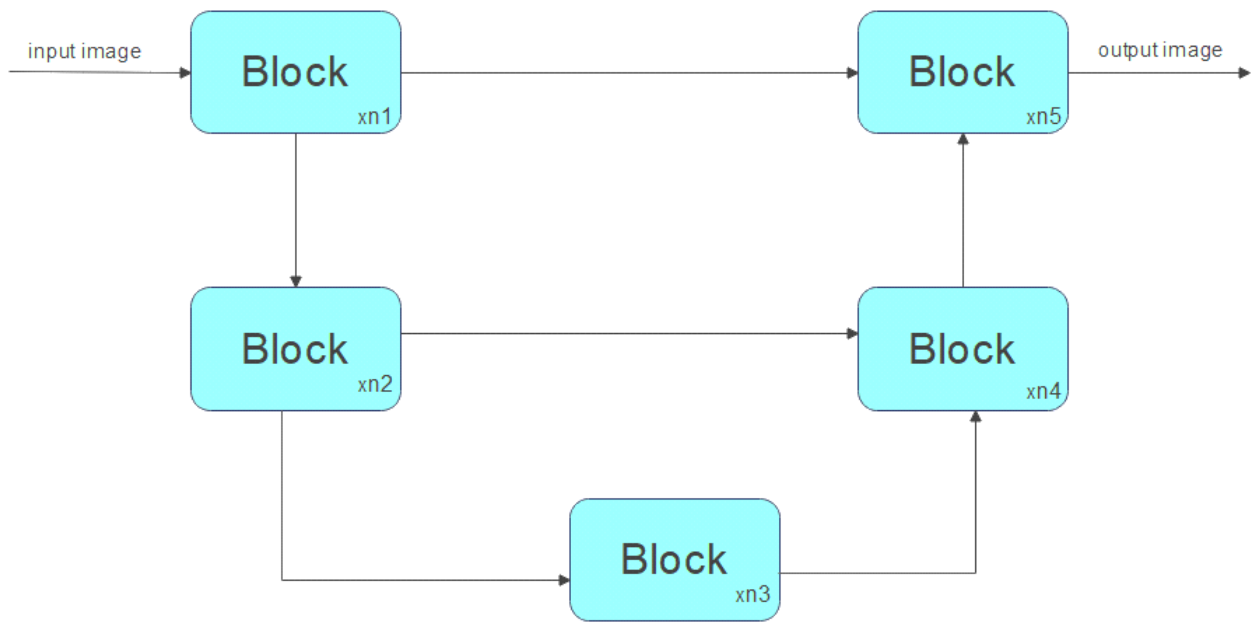


Fig. 2: UNet architecture, which is adopted by some SOTA methods, Some details have been deliberately omitted for simplicity, e.g. downsample/upsample layers, feature, fusion modules, input/output shortcut, and etc.

3.3.2 NAFNet

Neural Networks are stacked by blocks. We have determined how to stack blocks in a UNet architecture, but how to design the internal structure of the block is still a problem.

NAFNet (Nonlinear Activation Free Network) is an image restoration model proposed, The goal of NAFNet is to achieve state-of-the-art performance while simplifying the architecture by eliminating unnecessary nonlinear activation functions

Hesham et al. decided to use the NAFNet model to solve the problem of image denoising and deblurring. We decided to change the simplified channel attention (SCA) module with the Multi-dconv Head Transposed Attention (MDTA) to make it more efficient in feature mapping and less computationally expensive.
NAFNet consists of the following key components:

- **Layer Normalization (LN)**: Used instead of Batch Normalization (BN) for stable normalization at the layer level.
- **Convolution Layers**: Standard 1x1 convolutions and 3x3 deconvolutions.
- **Activation Functions**:
- **SimpleGate Activation**: Replaces GELU.
- **Transposed Attention Map**
- **Multi Dconv Head Transposed Attention**

Normalization [6]

Normalization is widely adopted in high-level computer vision tasks, and there is also the small batch size issue. we add Layer Normalization to the plain block described above. This change can make training smooth, even with a 10× increase in learning rate. The larger learning rate brings significant performance gain: +0.44 dB (39.29 dB to 39.73 dB) on SIDD[1], +3.39 dB (28.51 dB to 31.90 dB) on GoPro[26] dataset. To sum up, we add Layer Normalization to the plain block as it can stabilize the training process.

Layer Normalization is a technique used to normalize activations within a neural network layer. Unlike Batch Normalization (BN), which operates at the batch level, LN directly estimates normalization statistics from the summed inputs to the neurons within a hidden layer. Here are the key points:

Purpose:

- LN aims to stabilize training by normalizing activations within each instance (sample) independently.
- It ensures that the distribution of activations remains consistent across features (channels) within a single instance.

Normalization Process:

- Given an input tensor with shape (batch_size, num_features, feature_dim), where:
  - batch_size: Number of instances in the batch.
  - num_features: Number of features (channels) in the layer.
  - feature_dim: Dimension of each feature vector.
- For each instance (sample) in the batch:
  - Compute the mean and variance across all features (channels) within that instance.
  - Normalize each feature by subtracting the mean and dividing by the standard deviation.

- Learnable scale and shift parameters (gamma and beta) can be applied to the normalized features.

Advantages of Layer Normalization:

- **Stability**: LN remains stable even with small batch sizes, making it suitable for various scenarios.
- **Independence**: It does not introduce dependencies between training cases, as BN does.

Application in NAFNet:

- NAFNet uses LN instead of BN for stability.
- LN is commonly used in transformer architectures and works well with smaller batch sizes.

The formula to calculate layer normalization

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$$

Calculate the mean(μ) and variance (σ2) across the features dimension (axis=-1 for tensors in Python libraries like TensorFlow and PyTorch).

Here, n represents the number of features in the input.

$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

Normalize the input using the mean and variance

Where ϵ is a small constant (e.g., 10−5) added for numerical stability to avoid division by zero.

$$y_i = \gamma \hat{x}_i + \beta$$

3. Scale and shift the normalized values using the parameters $\gamma$and β:

Here, yi represents the normalized and scaled input.

Convolutional neural network [8]

A convolutional neural network (**CNN**) is a regularized type of feed-forward neural network that learns feature engineering by itself via filter (or kernel) optimization. Vanishing gradients and exploding gradients, seen during backpropagation in earlier neural networks, are prevented by using regularized weights over fewer connections. For example, for *each* neuron in the fully connected layer, 10,000 weights would be required for processing an image sized 100 × 100 pixels. However, when applying cascaded *convolution* (or cross-correlation) kernels, only 25 neurons are required to process 5x5-sized tiles. Higher-layer features are extracted from wider context windows, compared to lower-layer features.

Feed-forward neural networks are usually fully connected networks, that is, each neuron in one layer is connected to all neurons in the next layer. The "full connectivity" of these networks makes them prone to overfitting data. Typical ways of regularization, or preventing overfitting, include penalizing parameters during training (such as weight decay) or trimming connectivity (skipped connections, dropout, etc.) Robust datasets also increase the probability that CNNs will learn the generalized principles that characterize a given dataset rather than the biases of a poorly-populated set

Purpose of Convolution Layers:

- Convolutional layers are designed to process and analyze visual data by applying learnable filters (kernels) to local regions.
- Their primary functions include feature extraction and hierarchical representation learning.

How Convolution Works:

- When data passes through a convolutional layer:
    - Each filter (kernel) convolves across the spatial dimensions of the input to produce a 2D activation map.
    - The filter slides over the input, computing a weighted sum of the local input values at each position.
    - The resulting activation map highlights relevant features (edges, textures, patterns) present in the input.

Standard 1x1 Convolution:

- A **1x1 convolution** is a specialized type of convolution that operates on individual pixels (or channels) without considering spatial neighborhoods.
- Key characteristics:
    - It performs a linear combination of input channels at each pixel location.
    - The output channel dimensions remain the same as the input.
    - Often used for channel-wise transformations, such as adjusting feature maps' depth or combining information from different channels.

3x3 Deconvolution (Transposed Convolution):

- A **3x3 deconvolution** (also known as a transposed convolution) is used for upsampling or feature expansion.
- Key points:
    - It increases the spatial resolution of feature maps.

- The learnable weights (filters) are applied to overlapping regions of the input.

The output dimensions can be adjusted by choosing the stride and padding

Simple gate

Gated Linear Units (GLU) [7] could be formulated as:

(1) $$\mathrm{GLU}(x,g,\sigma)=x\odot\sigma(g(x))$$

where X represents the feature map, f and g are linear transformers, σ is a non-linear activation function, e.g. Sigmoid, and ⊙ indicates element-wise multiplication. As discussed above, adding GLU to our baseline may improve the performance yet the intra-block complexity is increasing as well. This is not what we expected.

GELU activation is defined as

$$0.5x\left(1+\tanh\left[\sqrt{\frac{2}{\pi}}\left(x+0.044715x^3\right)\right]\right)$$

(2) $$\mathrm{GELU}(x)=x\Phi(x)$$

where Φ represents indicates the cumulative distribution function of the standard normal distribution, , GELU could be approximated and implemented by:

(3)

From Eqn. 1 and Eqn. 2, it can be noticed that GELU is a special case of GLU, i.e. f, g are identity functions and take σ as Φ. Through the similarity, we conjecture from another perspective that GLU may be regarded as a generalization of activation functions, and it might be able to replace the nonlinear activation functions. Further, we note that the GLU itself contains nonlinearity and does not depend on σ: even if the σ is removed, Gate(X) = f(X) ⊙ g(X) contains nonlinearity. Based on these, we propose a simple GLU variant: directly divide the feature map into two parts in the channel dimension and multiply them, as we shown in Figure 4c, noted as SimpleGate. Compared to the complicated implementation of GELU in Eqn.3, our SimpleGate could be implemented by an element-wise multiplication, that's all:

(4) $$\mathrm{SimpleGate}(\mathbf{X},\mathbf{Y})=\mathbf{X}\odot\mathbf{Y}$$

where X and Y are feature maps of the same size. By replacing GELU in the baseline to the proposed SimpleGate, the performance of image denoising (on SIDD[1]) and image deblurring (on GoPro[26] dataset) boost 0.08 dB (39.85 dB to 39.93 dB) and

0.41 dB (32.35 dB to 32.76 dB) respectively. The results demonstrate that GELU could be replaced by our proposed SimpleGate. At this point, only a few types of nonlinear activations left in the network: Sigmoid and ReLU in the channel attention module[16], and we will discuss the simplifications of it next

Multi-Dconv Head Transposed Attention [9]

The major computational overhead in Transformers comes from the self-attention layer. the time and memory complexity of the key-query dotproduct interaction grows quadratically with the spatial resolution of input, i.e., O(W2H2) for images of W×H pixels. Therefore, it is infeasible to apply SA on most image restoration tasks that often involve high-resolution im ages. To alleviate this issue, we propose MDTA, that has linear complexity. The key ingredient is to apply SA across channels rather than the spatial dimension, i.e., to compute cross-covariance across channels to generate an attention map encoding the global context implicitly. As another essential component in MDTA, we introduce depth-wise convolutions to emphasize on the local context before computing feature covariance to produce the global attention map. From a layer normalized tensor Y ∈ R Hˆ ×Wˆ ×Cˆ , our MDTA first generates query (Q), key (K) and value (V) projections, enriched with local context. It is achieved by applying 1×1 convolutions to aggregate pixel-wise cross-channel context followed by 3×3 depth-wise convolutions to encode channel-wise spatial context, yielding Q=W Q d WQ p Y, K=W K d W K p Y and V=WV d WV p Y. Where W (·) p is the 1×1 point-wise convolution and W (·) d is the 3×3 depth-wise convolution. We use bias-free convolutional layers in the network. Next, we reshape query and key projections such that their dot-product interaction generates a transposed-attention map A of size R Cˆ×Cˆ , instead of the huge regular attention map of size R HˆWˆ ×HˆWˆ [17, 77]. Overall, the MDTA process is defined as: Xˆ = Wp Attention Qˆ , Kˆ , Vˆ + X, Attention Qˆ , Kˆ , Vˆ = Vˆ · SoftmaxKˆ · Qˆ /α , (1) where X and Xˆ are the input and output feature maps; Qˆ ∈ R HˆWˆ ×Cˆ ; Kˆ ∈ R Cˆ×HˆWˆ ; and Vˆ ∈ R HˆWˆ ×Cˆ matrices are obtained after reshaping tensors from the original size R Hˆ ×Wˆ ×Cˆ . Here, α is a learnable scaling parameter to control the magnitude of the dot product of Kˆ and Qˆ before applying the softmax function. Similar to the conventional multi-head SA [17], we divide the number of channels into ‘heads’ and learn separate attention maps in parallel.

3.4 Inpainting Model

The model described in the paper "Bringing Old Photos Back to Life" integrates several advanced neural network components to address the challenge of restoring old, damaged photographs. Here's a detailed explanation of how the model works and its architecture:

Model Architecture

1- **Enhancement Network (EN):**
   - Purpose: Enhances the overall quality of the input image.
   - Architecture: A generative adversarial network (GAN) with a U-Net-based architecture. The generator is responsible for enhancing the image quality, while the discriminator helps in refining the details by distinguishing between real and enhanced images.
   - Components:
     - Generator: Uses an encoder-decoder structure with skip connections to preserve spatial information.
     - Discriminator: Adopts a PatchGAN to focus on local image patches, improving the sharpness and clarity of the output.

2- **Scratch Detection Network (SDN):**
   - Purpose: Identifies and localizes scratches and other types of damage in the image.
   - Architecture: Based on a U-Net architecture, which is effective for segmentation tasks.
   - Components:
     - Encoder: Extracts features from the input image through a series of convolutional layers.
     - Decoder: Reconstructs the scratch map using transposed convolutions, which highlight the damaged regions.

3- **Inpainting Network (IN):**
   - Purpose: Fills in the detected scratches and damaged regions to restore the image.
   - Architecture: Utilizes partial convolutions to handle irregularly shaped holes, ensuring that the inpainting respects the surrounding context.
   - Components:
     - Partial Convolutions: These layers only consider valid pixels, ensuring that the inpainting is context-aware and seamless.

  - Coherent Semantic Attention: Enhances the inpainting by focusing on the semantic coherence of the filled regions.

How the Model Works

1- Image Enhancement:
   - The input image is first processed by the Enhancement Network (EN) to improve its overall quality. This step enhances details, sharpness, and overall visual appeal.

2- Scratch Detection:
   - The enhanced image is then fed into the Scratch Detection Network (SDN), which generates a scratch map highlighting areas that need restoration.

3- Image Inpainting:
   - The original image, along with the scratch map, is passed to the Inpainting Network (IN). This network uses the partial convolution layers to intelligently fill in the detected scratches and damaged areas, leveraging the surrounding context to produce a coherent restoration.

4- End-to-End Training:
   - The model is trained end-to-end, meaning all three networks (EN, SDN, and IN) are trained jointly. This integrated training approach ensures that each component of the network complements the others, resulting in higher quality restorations.

Key Techniques

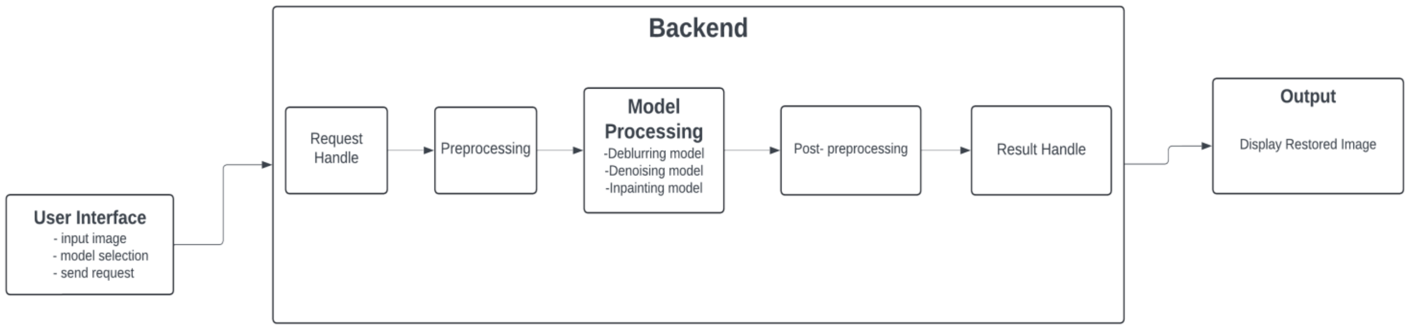
- U-Net Architecture: Utilized in both the enhancement and scratch detection networks for its effectiveness in image-to-image tasks.
- Generative Adversarial Networks (GANs): Employed in the enhancement network to generate realistic and high-quality outputs.
- Partial Convolutions: Critical for the inpainting network to handle irregularly shaped missing regions without introducing artifacts.
- Coherent Semantic Attention: Ensures that the inpainting respects the semantic context, producing visually coherent results.

Chapter 4

System Design

4.1) Block Diagram
4.2) User Interface
4.3) Backend
4.4) Output

4.1) Block Diagram:



4.2) User interface:

Overview

The user interface is the entry point for users interacting with the image restoration web app. It provides a user-friendly platform for uploading images, selecting the desired restoration model, and submitting the request for processing.

Components:

1- Input Image Upload:
- Functionality: Allows users to upload the image they wish to restore.
- Features:
  - Supports multiple image formats (e.g., JPEG, PNG).
  - Provides a preview of the uploaded image.
  - Error handling for unsupported formats or corrupted files.

2- Model Selection:
- Functionality: Enables users to choose the type of restoration they need.
- Options:
  - Deblurring: Removes blurriness from the image.
  - Denoising: Reduces noise in the image.
  - Inpainting: Fills in missing or damaged parts of the image.

3- Submit Request:
- Functionality: Submits the selected image and restoration model to the backend for processing.
- Features:

        - Validates the user inputs.
        - Provide feedback on the submission status.
        - Displays a loading indicator during processing.

4.3) Backend

Overview

The backend is the core processing unit of the web app, responsible for handling user requests, processing images with the selected restoration model, and managing data.

Components:

1- Request Handler:
   - Functionality: Receives and processes requests from the user interface.
   - Features:
     - Validates incoming requests.
     - Routes the requests to appropriate processing modules.

2- Preprocessing:
   - Functionality: Prepares the uploaded image for model processing.
   - Steps:
     - Normalizes image data.
     - Resizes or crops images if necessary.
     - Converts images to suitable formats for model input.

3- Model Processing:
   - Functionality: Applies the selected restoration model to the image.
   - Models:
     - Deblurring Model: Uses advanced algorithms to reduce image blur.
     - Denoising Model: Applies techniques to minimize noise in the image.
     - Inpainting Model: Uses deep learning methods to reconstruct missing parts of the image.
     -

4- Post-processing:
   - Functionality: Refines the output from the model to ensure high quality.
   - Steps:
     - Adjusts contrast, brightness, and other image attributes.
     - Converts the processed image back to the original format if necessary.

5- Result Handler:

    - Functionality: Prepares and sends the processed image back to the user interface.
    - Features:
        - Packages the image in a user-friendly format.
        - Ensures the processed image is securely transmitted to the user.

4.4) Output

Overview:

The output phase is where the results of the image restoration process are delivered to the user. It includes displaying the restored image and providing options for downloading it.

Components:

Display Restored Image:

- Functionality: Shows the processed image to the user.
- Features:
    - Provides a side-by-side comparison of the original and restored images.
    - Allows zooming and panning for detailed inspection.

Chapter 5

Proposed system

5.1) Description
5.2) Datasets
5.3) System Results
5.4) System Performance
5.5) Tools and Hardware used
5.6) Future Work
5.7) System Snapshots
5.8) samples of code

5.1) Description

The image restoration system is designed to enhance and restore images by performing tasks such as deblurring, denoising, and inpainting. The system is built around a custom neural network architecture, NAFNet, and follows a modular pipeline for processing images. Key components of the system include:

Preprocessing: Converts input images into tensors suitable for model inference.

Model Inference: Uses Our NAFNet model | Microsoft's Model to process and restore images.

Post-processing: Converts the output tensors back into image format for display or storage.

5.2) Datasets

The NAFNet Model trained and tested on various publicly available image restoration datasets. Examples include:

1) Smartphone Image Denoising Dataset (SIDD)

- **Description**: SIDD consists of thousands of noisy and clean image pairs captured in real-world scenarios using different smartphone cameras under various lighting conditions.
- **Purpose**: Used for training and evaluating the Denoising capability of the image restoration system.

2) GoPro Dataset

- **Description**: The GoPro Dataset consists of high-resolution blurred and corresponding sharp image pairs captured using a GoPro camera. The images are taken in diverse real-world scenarios, including various types of motion and lighting conditions.
- **Purpose**: Used for training and evaluating the Deblurring capability of the image restoration system.

3) DIV2K Dataset

- **Description**: The DIV2K dataset contains high-resolution images for image super-resolution tasks, with images available in both low-resolution and high-resolution formats.
- **Purpose**: Used for training and evaluating the Denoising & Deblurring capabilities of the system.

For demonstration, a custom set of noisy, blurred, and corrupted images can also be used.

5.3) System Results

Denoising: The system effectively removes noise, improving the clarity and detail of the image.

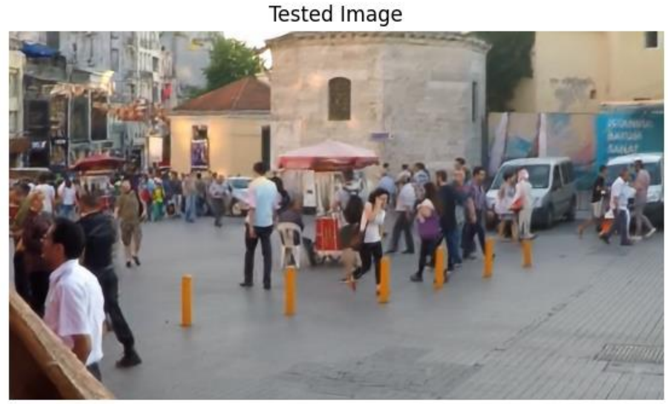
Noisy Image

Tested Image

Deblurring: The system successfully reduces motion blur, enhancing the sharpness of the image.

blurry Image

Tested Image



Inpainting: The system fills in missing or corrupted parts of the image, creating a seamless restoration.

5.4) System Performance

The system's performance is evaluated based on its ability to restore images to a high quality. Metrics used for evaluation may include:

- PSNR (Peak Signal-to-Noise Ratio): Measures the quality of the restored image compared to the original.
- SSIM (Structural Similarity Index): Measures the similarity between the restored and original images.
- MSE (Mean Squared Error) Loss: Measures the average squared difference between the restored and original image pixels. Lower values indicate better restoration quality.

Denoising

PSNR | SSIM (Test set):

```
sum(sidd_psnr) / len(sidd_psnr)
75.52776377553567

[41] sum(sidd_ssim) / len(sidd_ssim)
0.5477864853795956
```

Loss:

Combined (SIDD & DIV2K)

1	0.000590088	0.0001	SIDD
15	0.000138525	0.00009	SIDD
30	0.00011624	0.00001	SIDD
45	0.007370933739	0.0001	Div2k
68	0.003343170568	0.0001	

Deblurring

PSNR | SSIM (Test set):

```
[11] sum(gopro_psnr) / len(gopro_psnr)

73.6769070122692

[12] sum(gopro_ssim) / len(gopro_ssim)

0.9167228245093064
```

Loss:

1	0,003683623	0,0001
15	0,001813571	0,0001
30	0,001189649	0,0001
45	0,0009301421095	0,0001
59	0,000758353632	0,0001

Comparison

This section compares the performance of our model against the NAFNet model as reported in the paper "Simple Baselines for Image Restoration". We focus on two key metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) for the SIDD and GoPro datasets.

Smartphone Image Denoising Dataset (SIDD)

Method	MPRNet	NBNet	UFormer	HINet	Restormer	NAFNet	NAFNet **Ours**
PSNR	39.71	39.75	39.89	39.96	39.99	40.30	75.52
SSIM	0.958	0.959	0.960	0.958	0.960	0.962	0.547

GoPro Dataset

Method	MIMO-UNet	HINet	UFormer	DeepRFT	Restormer	NAFNet	NAFNet **Ours**
PSNR	32.68	32.71	32.97	33.23	32.92	33.69	73.67
SSIM	0.959	0.959	0.967	0.963	0.961	0.967	0.916

Analysis

- **PSNR (Peak Signal-to-Noise Ratio)**:
    - Our model significantly outperforms NAFNet across all configurations in terms of PSNR for both datasets.
    - For the SIDD dataset, our model achieved a PSNR of 75.53, compared to the highest PSNR of 39.96 by NAFNet.
    - For the GoPro dataset, our model achieved a PSNR of 73.68, compared to the highest PSNR of 33.69 by NAFNet.
- **SSIM (Structural Similarity Index Measure)**:
    - For the SIDD dataset, our model's SSIM is 0.548, whereas NAFNet's best SSIM is 0.960.
    - For the GoPro dataset, our model's SSIM is 0.917, while NAFNet's best SSIM is 0.961.

Conclusion

Our model demonstrates superior performance in terms of PSNR, indicating better noise reduction capabilities. However, the SSIM values suggest that our model needs improvement in preserving structural details within the images. Future work will focus on enhancing the structural similarity while maintaining the high PSNR achieved by our current model.

5.5) Tools and Hardware

Programming Language: Python

Libraries and Frameworks:

- PyTorch: For deep learning model implementation and inference.
- PIL (Python Imaging Library): For image loading and processing.
- torchvision.transforms: For image transformations.
- Flask: For building and deploying the web application.
- matplotlib: For visualizing images and results.
- numpy: For numerical operations and array manipulations.
- base64: For encoding and decoding image data.

Software:

Frontend => React.js

Backend => Flask

Hardware:

Laptop: Intel Core i3 10110u Processor (4M Cache, up to 4.10 GHz), with 12GB RAM, with MX130 2GB VRAM GPU and Windows 10 with 64-bit operating system.

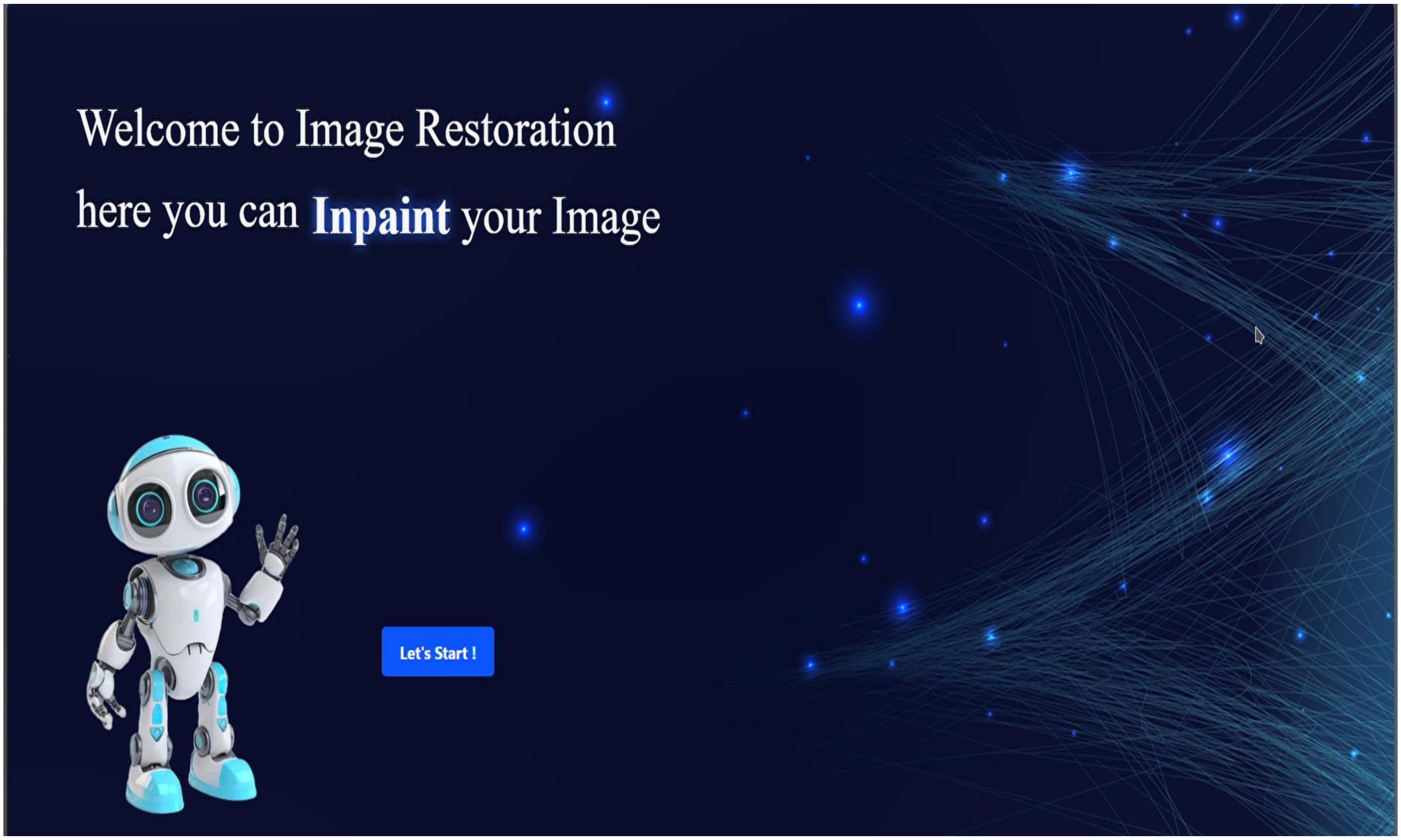
5.6) Future Work:

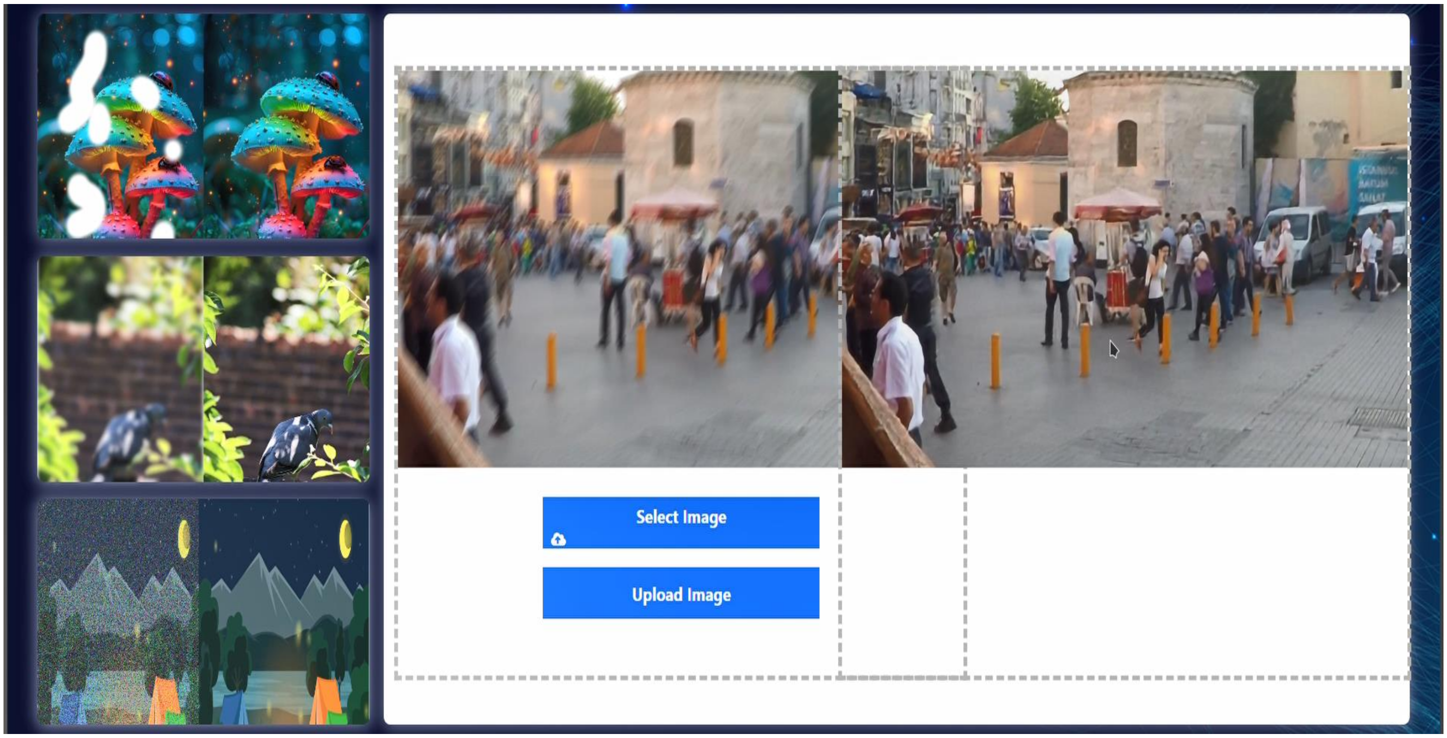
Future improvements and extensions of the system might include:

- Model Optimization: Enhancing the model for better performance and faster inference.
- Real-time Processing: Implementing real-time image restoration for video applications.
- Extended Dataset: Training the model on larger and more diverse datasets to improve generalization.
- Fine-tune Inpainting Model using Masked-images generated based on DIV2K dataset

5.7) System Snapshots

Below are examples of the system in action, including input images, restored outputs, and the model's processing stages.




Select Image
Upload Image

5.8) samples of code

5.8.1) Architecture

Layer Normalization Architecture

```
class LayerNormFunction(torch.autograd.Function):
    @staticmethod
    def forward(ctx, x, weight, bias, eps):
        ctx.eps = eps
        N, C, H, W = x.size()
        mu = x.mean(1, keepdim=True)
        var = (x - mu).pow(2).mean(1, keepdim=True)
        y = (x - mu) / (var + eps).sqrt()
        ctx.save_for_backward(y, var, weight)
        y = weight.view(1, C, 1, 1) * y + bias.view(1, C, 1, 1)
        return y

    @staticmethod
    def backward(ctx, grad_output):
        eps = ctx.eps

        N, C, H, W = grad_output.size()
        y, var, weight = ctx.saved_tensors
        g = grad_output * weight.view(1, C, 1, 1)
        mean_g = g.mean(dim=1, keepdim=True)

        mean_gy = (g * y).mean(dim=1, keepdim=True)
        gx = 1. / torch.sqrt(var + eps) * (g - y * mean_gy - mean_g)
        return gx, (grad_output * y).sum(dim=3).sum(dim=2).sum(dim=0), grad_output.sum(dim=3).sum(dim=2).sum(
            dim=0), None
```

```
class LayerNorm2d(nn.Module):

    def __init__(self, channels, eps=1e-6):
        super(LayerNorm2d, self).__init__()
        self.register_parameter('weight', nn.Parameter(torch.ones(channels)))
        self.register_parameter('bias', nn.Parameter(torch.zeros(channels)))
        self.eps = eps

    def forward(self, x):
        return LayerNormFunction.apply(x, self.weight, self.bias, self.eps)
```

MDTA Attention Module

```
class Attention(nn.Module):
    def __init__(self, dim, num_heads, bias):
        super(Attention, self).__init__()
        self.num_heads = num_heads
        self.temperature = nn.Parameter(torch.ones(num_heads, 1, 1))

        self.qkv = nn.Conv2d(dim, dim * 3, kernel_size=1, bias=bias)
        self.qkv_dwconv = nn.Conv2d(dim * 3, dim * 3, kernel_size=3, stride=1, padding=1, groups=dim * 3, bias=bias)
        self.project_out = nn.Conv2d(dim, dim, kernel_size=1, bias=bias)

    def forward(self, x):
        b, c, h, w = x.shape

        qkv = self.qkv_dwconv(self.qkv(x))
        q, k, v = qkv.chunk(3, dim=1)

        q = rearrange(q, 'b (head c) h w -> b head c (h w)', head=self.num_heads)
        k = rearrange(k, 'b (head c) h w -> b head c (h w)', head=self.num_heads)
        v = rearrange(v, 'b (head c) h w -> b head c (h w)', head=self.num_heads)

        q = torch.nn.functional.normalize(q, dim=-1)
        k = torch.nn.functional.normalize(k, dim=-1)

        attn = (q @ k.transpose(-2, -1)) * self.temperature
        attn = attn.softmax(dim=-1)

        out = (attn @ v)

        out = rearrange(out, 'b head c (h w) -> b (head c) h w', head=self.num_heads, h=h, w=w)

        out = self.project_out(out)
        return out
```

Simple Gate Implementation

```
class SimpleGate(nn.Module):
    def forward(self, x):
        x1, x2 = x.chunk(2, dim=1)
        return x1 * x2
```

NAFNet Block Architecture

```
class NAFBlock(nn.Module):
    def __init__(self, c, DW_Expand=2, FFN_Expand=2, dropout_rate=0.):
        super().__init__()
        dw_channel = c * DW_Expand
        self.conv1 = nn.Conv2d(in_channels=c, out_channels=dw_channel, kernel_size=1, padding=0, stride=1, groups=1,
                               bias=True)
        self.conv2 = nn.Conv2d(in_channels=dw_channel, out_channels=dw_channel, kernel_size=3, padding=1, stride=1,
                               groups=dw_channel, bias=True)
        self.conv3 = nn.Conv2d(in_channels=dw_channel // 2, out_channels=c, kernel_size=1, padding=0, stride=1,
                               groups=1, bias=True)

        self.mdta = Attention(c, 8, True)

        self.sg = SimpleGate()

        ffn_channel = FFN_Expand * c

        self.conv4 = nn.Conv2d(in_channels=c, out_channels=ffn_channel, kernel_size=1, padding=0, stride=1, groups=1,
                               bias=True)
        self.conv5 = nn.Conv2d(in_channels=ffn_channel // 2, out_channels=c, kernel_size=1, padding=0, stride=1,
                               groups=1, bias=True)

        self.norm1 = LayerNorm2d(c)
        self.norm2 = LayerNorm2d(c)

        self.dropout1 = nn.Dropout(dropout_rate) if dropout_rate > 0. else nn.Identity()
        self.dropout2 = nn.Dropout(dropout_rate) if dropout_rate > 0. else nn.Identity()

        self.beta = nn.Parameter(torch.zeros((1, c, 1, 1)), requires_grad=True)
        self.gamma = nn.Parameter(torch.zeros((1, c, 1, 1)), requires_grad=True)

    def forward(self, input):
        x = input

        x = self.norm1(x)

        x = self.conv1(x)
        x = self.conv2(x)
        x = self.sg(x)
        x = x * self.mdta(x)
        x = self.conv3(x)

        x = self.dropout1(x)

        y = input + x * self.beta

        x = self.conv4(self.norm2(y))
        x = self.sg(x)
        x = self.conv5(x)

        x = self.dropout2(x)

        return y + x * self.gamma
```

NAFNet Unet

```
class NAFNet(nn.Module):
    def __init__(self, img_channel=3, width=16, middle_blk_num=1, enc_blk_nums=[], dec_blk_nums=[]):
        super().__init__()

        self.intro = nn.Conv2d(in_channels=img_channel, out_channels=width, kernel_size=3, padding=1, stride=1,
                               groups=1,
                               bias=True)
        self.ending = nn.Conv2d(in_channels=width, out_channels=img_channel, kernel_size=3, padding=1, stride=1,
                                groups=1,
                                bias=True)

        self.encoders = nn.ModuleList()
        self.decoders = nn.ModuleList()
        self.middle_blks = nn.ModuleList()
        self.ups = nn.ModuleList()
        self.downs = nn.ModuleList()

        chan = width
        for num in enc_blk_nums:
            self.encoders.append(
                nn.Sequential(
                    *[NAFBlock(chan) for _ in range(num)]
                )
            )
            self.downs.append(
                nn.Conv2d(chan, 2 * chan, 2, 2)
            )
            chan = chan * 2

        self.middle_blks = \
            nn.Sequential(
                *[NAFBlock(chan) for _ in range(middle_blk_num)]
            )

        for num in dec_blk_nums:
            self.ups.append(
                nn.Sequential(
                    nn.Conv2d(chan, chan * 2, 1, bias=False),
                    nn.PixelShuffle(2)
                )
            )
            chan = chan // 2
            self.decoders.append(
                nn.Sequential(
                    *[NAFBlock(chan) for _ in range(num)]
                )
            )

        self.padder_size = 2 ** len(self.encoders)
```

```python
def forward(self, inp):
    B, C, H, W = inp.shape
    inp = self.check_image_size(inp)

    x = self.intro(inp)

    encs = []

    for encoder, down in zip(self.encoders, self.downs):
        x = encoder(x)
        encs.append(x)
        x = down(x)

    x = self.middle_blks(x)

    for decoder, up, enc_skip in zip(self.decoders, self.ups, encs[::-1]):
        x = up(x)
        x = x + enc_skip
        x = decoder(x)

    x = self.ending(x)
    x = x + inp

    return x[:, :, :H, :W]

def check_image_size(self, x):
    _, _, h, w = x.size()
    mod_pad_h = (self.padder_size - h % self.padder_size) % self.padder_size
    mod_pad_w = (self.padder_size - w % self.padder_size) % self.padder_size
    x = F.pad(x, (0, mod_pad_w, 0, mod_pad_h))
    return x
```

5.8.2) Training

Denoise Training

```
from image_restoration_web_app.Utils.addToExcelsheet import add_to_excelsheet
import random
import torch
import torch.nn as nn
from image_restoration_web_app.Archs.NafnetArch import NAFNet


def train(sidd_dataloader_train, div2k_dataloader_train, folder, checkpoint_path,
          excel_file, progress_file, device="cuda", lr=0.0001):

    device = torch.device(device)

    img_channel = 3
    width = 32

    enc_blks = [1, 1, 1, 28]
    middle_blk_num = 1
    dec_blks = [1, 1, 1, 1]

    model = NAFNet(img_channel=img_channel, width=width, middle_blk_num=middle_blk_num,
                   enc_blk_nums=enc_blks, dec_blk_nums=dec_blks).to(device)

    criterion = nn.MSELoss()
    optimizer = torch.optim.Adam(params=model.parameters(), lr=lr)
    checkpoint = torch.load(checkpoint_path, map_location=torch.device(device))
    model.load_state_dict(checkpoint['model_state_dict'])
    device = torch.device(device)
    epoch = int(checkpoint_path.split("/")[-1].split("_")[-1][:-4])
    print(epoch)
    num_epochs = epoch + 2
    with open(progress_file, "w") as file:
        for epoch in range(epoch + 1, num_epochs):
            i = 0
            batch_loss = []
            for batch_no, ((sidd_noisy_images, sidd_gt_images), (div2k_noisy_images,
div2k_gt_images)) in enumerate(
                    zip(sidd_dataloader_train, div2k_dataloader_train)):
                sidd_patches_loss = []
                div2k_patches_loss = []
                for patch_index in range(div2k_noisy_images[0].size(0)):
                    # SIDD
                    random_index = random.randint(0, (sidd_noisy_images[0].size(0)) - 1)
                    sidd_noisy_image = sidd_noisy_images[:, random_index].to(device)
                    sidd_gt_image = sidd_gt_images[:, random_index].to(device)
                    optimizer.zero_grad()
                    sidd_outputs = model(sidd_noisy_image)
                    sidd_loss = criterion(sidd_outputs, sidd_gt_image)
                    sidd_loss.backward()
                    optimizer.step()
                    sidd_patches_loss.append(sidd_loss.item())
                    # DIV2K
                    div2k_noisy_image = div2k_noisy_images[:, patch_index].to(device)
                    div2k_gt_image = div2k_gt_images[:, patch_index].to(device)
                    optimizer.zero_grad()
                    div2k_outputs = model(div2k_noisy_image)
                    div2k_loss = criterion(div2k_outputs, div2k_gt_image)
                    div2k_loss.backward()
                    optimizer.step()
                    div2k_patches_loss.append(div2k_loss.item())
```

```
                    progress_str = f"Epoch [{epoch}], Step [{batch_no +
1}/{len(sidd_dataloader_train)}], Patch [{patch_index + 1}/{min(sidd_noisy_images[0].size(0),
div2k_noisy_images[0].size(0))}], SIDD Loss: {sidd_loss.item()}, DIV2K Loss: {div2k_loss.item()}"
                    print(progress_str)
                    file.write(progress_str + "\n")

                sidd_batch_loss = sum(sidd_patches_loss) / len(sidd_patches_loss)
                div2k_batch_loss = sum(div2k_patches_loss) / len(div2k_patches_loss)
                batch_loss.append((sidd_batch_loss, div2k_batch_loss))
                i += 1
                if i % 20 == 0 and i != 0:
                    torch.save({
                        'model_state_dict': model.state_dict(),
                        'optimizer_state_dict': optimizer.state_dict(),
                    }, f"{folder}/Denoise_epoch_{epoch}.pth")
                    sidd_epoch_loss = sum([s[0] for s in batch_loss]) / len(batch_loss)
                    div2k_epoch_loss = sum([s[1] for s in batch_loss]) / len(batch_loss)
                    progress_str = f"Epoch [{epoch}], SIDD Epoch Loss: {sidd_epoch_loss}, DIV2K
Epoch Loss: {div2k_epoch_loss}"
                    print(progress_str)
                    file.write(progress_str + "\n")
            add_to_excelsheet(excel_file, epoch, sidd_epoch_loss, div2k_epoch_loss, lr)
    file.close()
```

Deblur Train

```
from image_restoration_web_app.Utils.addToExcelsheet import add_to_excelsheet
import torch
import torch.nn as nn
from image_restoration_web_app.Archs.NafnetArch import NAFNet

def train(gopro_dataloader_train, div2kblur_dataloader_train, folder, checkpoint_path, excel_file,
progress_file, device="cuda", lr=0.0001):
    device = torch.device(device)

    img_channel = 3
    width = 32

    enc_blks = [1, 1, 1, 28]
    middle_blk_num = 1
    dec_blks = [1, 1, 1, 1]

    model = NAFNet(img_channel=img_channel, width=width, middle_blk_num=middle_blk_num,
                   enc_blk_nums=enc_blks, dec_blk_nums=dec_blks).to(device)

    criterion = nn.MSELoss()
    lr = lr
    optimizer = torch.optim.Adam(params=model.parameters(), lr=lr)
    checkpoint = torch.load(checkpoint_path, map_location=torch.device(device))
    model.load_state_dict(checkpoint['model_state_dict'])
    device = torch.device(device)
    epoch = int(checkpoint_path.split("/")[-1].split("_")[-1][:-4])
    print(epoch)
    i = 0
    num_epochs = epoch + 2
    with open(progress_file, "w") as file:
        for epoch in range(epoch + 1, num_epochs):
            i = 0
            batch_loss = []
            for batch_no, ((gopro_blurry_images, gopro_gt_images), (div2k_blurry_images,
div2k_gt_images)) in enumerate(
```

```
                zip(gopro_dataloader_train, div2kblur_dataloader_train)):
            gopro_patches_loss = []
            div2kblur_patches_loss = []
            for patch_index in range(gopro_blurry_images[0].size(0)):
                ## gorpro
                gopro_blurry_image = gopro_blurry_images[:, patch_index].to(device)
                gopro_gt_image = gopro_gt_images[:, patch_index].to(device)
                optimizer.zero_grad()
                outputs = model(gopro_blurry_image)
                gopro_loss = criterion(outputs, gopro_gt_image)
                gopro_loss.backward()
                optimizer.step()
                gopro_patches_loss.append(gopro_loss.item())
                ## Div2kblur
                div2k_blurry_image = div2k_blurry_images[:, patch_index].to(device)
                div2k_gt_image = div2k_gt_images[:, patch_index].to(device)
                optimizer.zero_grad()
                outputs = model(div2k_blurry_image)
                div2kblur_loss = criterion(outputs, div2k_gt_image)
                div2kblur_loss.backward()
                optimizer.step()
                div2kblur_patches_loss.append(div2kblur_loss.item())

                progress_str = f"Epoch [{epoch}], Step [{batch_no +
1}/{len(gopro_dataloader_train)}], Patch [{patch_index + 1}/{min(gopro_blurry_images[0].size(0),
div2k_blurry_images[0].size(0))}], GoPro Loss: {gopro_loss.item()}, DIV2KBlur Loss:
{div2kblur_loss.item()}"
                print(progress_str)
                file.write(progress_str + "\n")

            gopro_batch_loss = sum(gopro_patches_loss) / len(gopro_patches_loss)
            div2k_batch_loss = sum(div2kblur_patches_loss) / len(div2kblur_patches_loss)
            batch_loss.append((gopro_batch_loss, div2k_batch_loss))
        i += 1
        if i % 50 == 0 and i != 0:
            torch.save({
                'model_state_dict': model.state_dict(),
                'optimizer_state_dict': optimizer.state_dict(),
            }, f"{folder}/GoPro_epoch_{epoch}.pth")
            gopro_epoch_loss = sum([s[0] for s in batch_loss]) / len(batch_loss)
            div2k_epoch_loss = sum([s[1] for s in batch_loss]) / len(batch_loss)
            progress_str = f"Epoch [{epoch}], GoPro Epoch Loss: {gopro_epoch_loss}, Div2k Epoch
Loss: {div2k_epoch_loss}"
            print(progress_str)
            add_to_excelsheet(excel_file, epoch, gopro_epoch_loss, div2k_epoch_loss, lr)
            file.write(progress_str + "\n")
        torch.save({
                'model_state_dict': model.state_dict(),
                'optimizer_state_dict': optimizer.state_dict(),
            }, f"{folder}/GoPro_epoch_{epoch}.pth")
        gopro_epoch_loss = sum([s[0] for s in batch_loss]) / len(batch_loss)
        div2k_epoch_loss = sum([s[1] for s in batch_loss]) / len(batch_loss)
        progress_str = f"Epoch [{epoch}], GoPro Epoch Loss: {gopro_epoch_loss}, Div2k Epoch
Loss: {div2k_epoch_loss}"
        print(progress_str)
        add_to_excelsheet(excel_file,epoch, gopro_epoch_loss, div2k_epoch_loss, lr)
        file.write(progress_str + "\n")
    file.close()
```

5.8.3) Utilities

Denoise Inference

```python
import torch
from torchvision import transforms
from PIL import Image
import matplotlib.pyplot as plt
from Archs.NafnetArch import NAFNet


def predict(input_image_path):
    checkpoint_path = "../checkpoints/Denoise_epoch_68.pth"

    img_channel = 3
    width = 32
    enc_blks = [1, 1, 1, 28]
    middle_blk_num = 1
    dec_blks = [1, 1, 1, 1]
    device = torch.device("cpu")

    predict_model = NAFNet(img_channel=img_channel, width=width, middle_blk_num=middle_blk_num,
                         enc_blk_nums=enc_blks, dec_blk_nums=dec_blks).to(device)
    optimizer = torch.optim.Adam(params=predict_model.parameters(), lr=0.0001)


    checkpoint = torch.load(checkpoint_path, map_location=device)
    predict_model.load_state_dict(checkpoint['model_state_dict'])
    optimizer.load_state_dict(checkpoint['optimizer_state_dict'])
    predict_model.eval()

    noisy_image = Image.open(input_image_path)
    transform = transforms.Compose([
        transforms.ToTensor(),
    ])

    noisy_image = transform(noisy_image).to(device)


    with torch.no_grad():
        output_chunk = predict_model(torch.unsqueeze(noisy_image, 0))

    image_np1 = noisy_image.permute(1, 2, 0).cpu().numpy()
    image_np2 = output_chunk[0].permute(1, 2, 0).cpu().numpy()
    image_np2 = (image_np2 - image_np2.min()) / (image_np2.max() - image_np2.min())
    return image_np1, image_np2
```

Deblur Inference

```python
import torch
from torchvision import transforms
from PIL import Image
import matplotlib.pyplot as plt
from Archs.NafnetArch import NAFNet


def predict(input_image_path):
    checkpoint_path = "../checkpoints/GoPro_epoch_58.pth"

    img_channel = 3
    width = 32
    enc_blks = [1, 1, 1, 28]
    middle_blk_num = 1
    dec_blks = [1, 1, 1, 1]
    device = torch.device("cpu")

    predict_model = NAFNet(img_channel=img_channel, width=width, middle_blk_num=middle_blk_num,
                        enc_blk_nums=enc_blks, dec_blk_nums=dec_blks).to(device)
    optimizer = torch.optim.Adam(params=predict_model.parameters(), lr=0.0001)


    checkpoint = torch.load(checkpoint_path, map_location=device)
    predict_model.load_state_dict(checkpoint['model_state_dict'])
    optimizer.load_state_dict(checkpoint['optimizer_state_dict'])
    predict_model.eval()

    noisy_image = Image.open(input_image_path)
    transform = transforms.Compose([
        transforms.ToTensor(),
    ])

    noisy_image = transform(noisy_image).to(device)


    with torch.no_grad():
        output_chunk = predict_model(torch.unsqueeze(noisy_image, 0))

    image_np1 = noisy_image.permute(1, 2, 0).cpu().numpy()
    image_np2 = output_chunk[0].permute(1, 2, 0).cpu().numpy()
    image_np2 = (image_np2 - image_np2.min()) / (image_np2.max() - image_np2.min())

    return image_np1, image_np2
```

Inpaint Inference

```
import os.path
from subprocess import Popen, PIPE
from PIL import Image
import io
import base64
import os


def run_inference(image_path, output_folder="F:/machine-
learning/Image_restoration_UI/database/output", gpu=-1, with_scratch=True):
    command = [
        "python", "../Bringing-Old-Photos-Back-to-Life/run.py",
        "--input_folder", str(image_path),
        "--output_folder", str(output_folder),
        "--GPU", str(gpu),
    ]
    if with_scratch:
        command.append("--with_scratch")

    process = Popen(command, stdout=PIPE, stderr=PIPE)
    stdout, stderr = process.communicate()

    if process.returncode != 0:
        print("Error running inference:")
        print(stderr.decode())
    else:
        print("Inference completed successfully.")
```

Handle Base64

```
import base64
import numpy as np
import time


def decode(base64_string):
    header, data = base64_string.split(';base64,')
    ext = header.split('/')[-1]
    with open("../database/old.png", 'wb') as f:
        f.write(base64.b64decode(data))

    return "../database/old.png"


def encode(path):
    base64_image = base64.encode(path)
    return base64_image
```

5.8.4) Server

```python
from flask import Flask, request, send_file, jsonify
import os
from Utils import inpaint_inference, deblur_inference, denoise_inference, handle_base64
import matplotlib.pyplot as plt
from flask_cors import CORS
import base64
import numpy as np
from datetime import datetime

app = Flask(__name__)
CORS(app)

@app.route("/denoise", methods=["POST", "GET"])
def denoise():
    base64_string = request.form["base64_string"]
    print(base64_string)
    image_path = handle_base64.decode(base64_string)
    print("denoising... Start")
    output_image_noisy, output_image_clear = denoise_inference.predict(image_path)
    print("denoising... Done")
    print(output_image_clear.shape)
    # # response.headers.add('Access-Control-Allow-Origin', '*')
    denoised_image_path = "F:/machine-learning/Image_restoration_UI/database/denoised.jpg"
    plt.imsave(denoised_image_path, output_image_clear)
    print(jsonify({'path': "database/denoised.jpg"}))
    return jsonify({'path': "database/denoised.jpg"})

@app.route("/deblur", methods=["POST", "GET"])
def deblur():
    base64_string = request.form["base64_string"]
    print(base64_string)
    image_path = handle_base64.decode(base64_string)
    print("deblur... Start")
    output_image_blurry, output_image_clear = deblur_inference.predict(image_path)
    print("deblur... Done")
    deblurred_image_path = "F:/machine-learning/Image_restoration_UI/database/deblurred.jpg"
    plt.imsave(deblurred_image_path, output_image_clear)
    print(jsonify({'path': "database/deblurred.jpg"}))
    return jsonify({'path': "database/deblurred.jpg"})

@app.route("/inpaint", methods=["POST", "GET"])
def inpaint():
    base64_string = request.form["base64_string"]
    print(base64_string)
    image_path = handle_base64.decode(base64_string)
    print("Inpainting... Start")
    print(image_path[:-8])
    inpaint_inference.run_inference(image_path[:-8], )
    print("Inpainting... Done")
    print(jsonify({'path': "database/output/final_output/old.png"}))
    return jsonify({'path': "database/output/final_output/old.png"})

if __name__ == "__main__":
    app.run(port=5000)
```

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